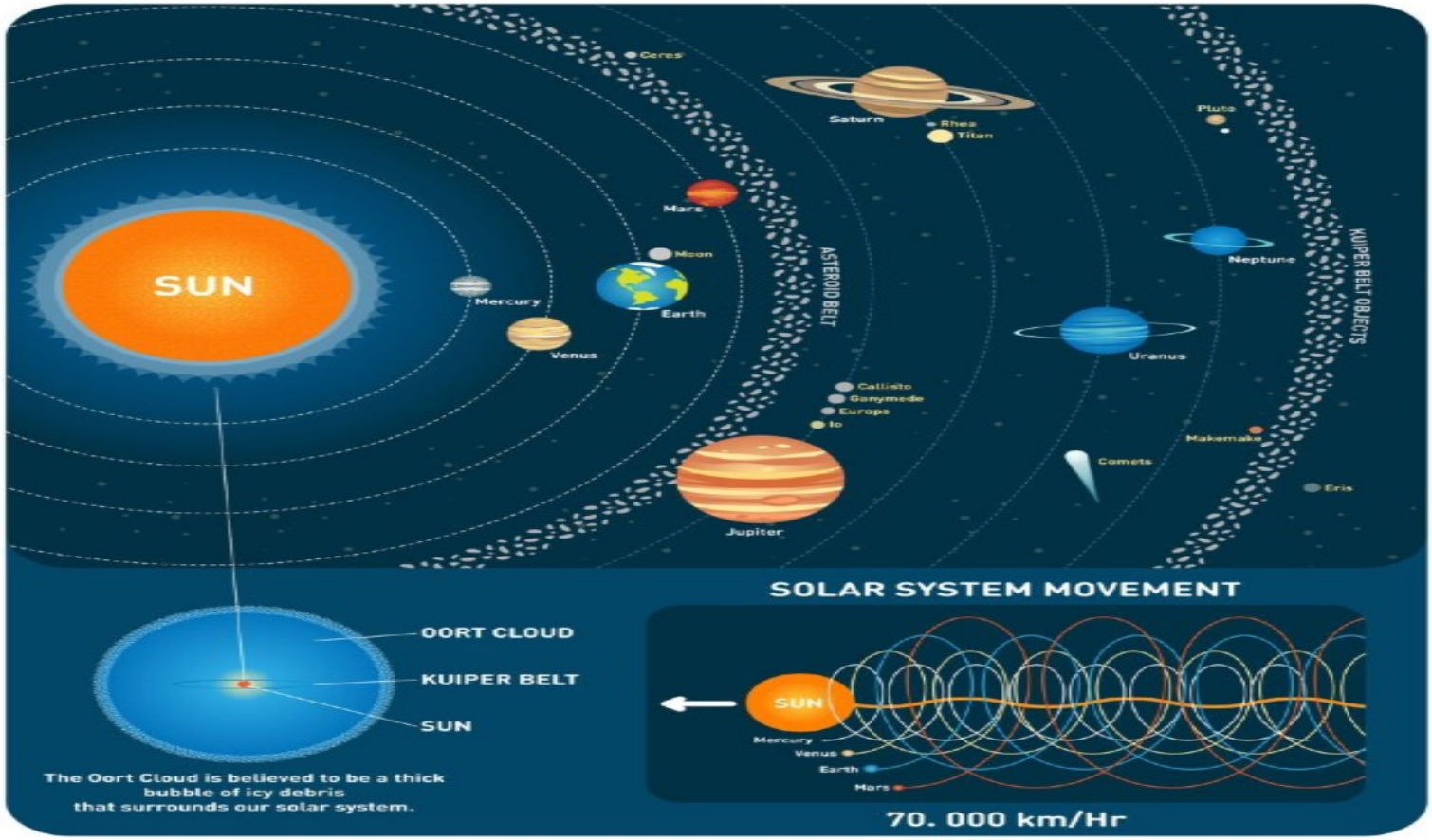
II.1. Introduction The Sun is a source of light that illuminates the Earth and the other components of the solar system. To the naked eye, the light emitted by the Sun appears white, but if this light is separated under the right conditions, all the colors of the rainbow are revealed. We will see later in this chapter that, to describe the arrangement of electrons around an atom's nucleus, we must analyze the light absorbed and emitted by atoms, which requires some understanding of the nature of light



II.2. The Wave Nature of Light

II.2.1. Electromagnetic Waves :An electromagnetic wave consists of both an electric field 'E' and a magnetic field 'B' oscillating at the same frequency (ν, nu). These two fields, which are perpendicular to each other, propagate through a medium in orthogonal directions .

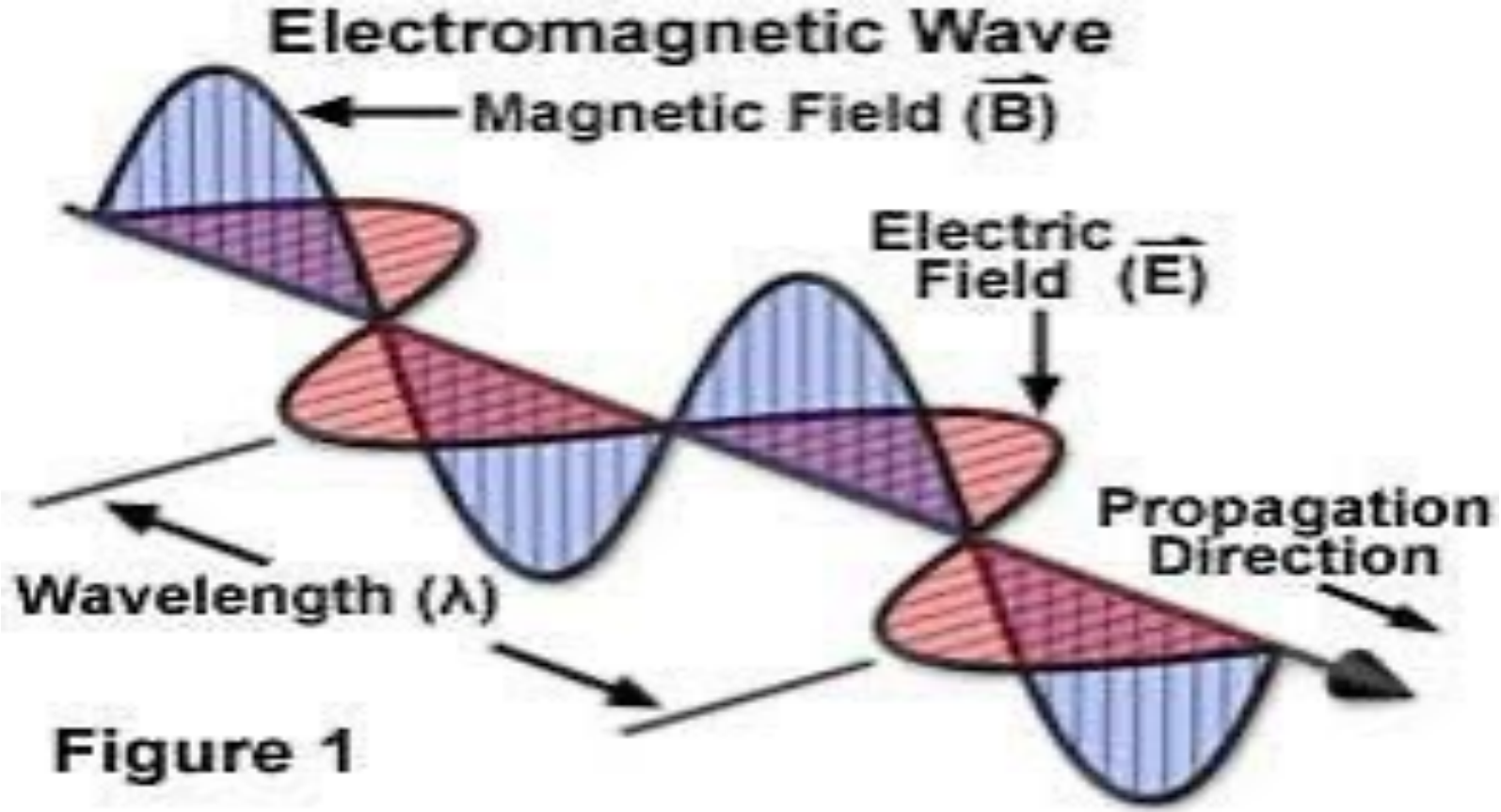


Figure 1

These waves propagate at a speed that depends on the medium in question.
In a vacuum, the speed is equal to $3 \cdot 10^8$ $m \cdot s^{-1}$.
A wave is characterized by several physical quantities :
The wavelength (λ, lambda) is the length of one cycle of a wave, the distance between two successive peaks. It is measured in meters or one of its submultiples:
- The nanometer (1 nm = 10^{-9} m)
- The micrometer (1 µm = 10^{-6} m)
- The centimeter (1 cm = 10^{-2} m)

The period T represents the time required for the wave to complete one cycle. The unit is the second.

The frequency ν is the inverse of the period; it represents the number of cycles per unit of time. It is expressed in hertz (Hz): $\nu = 1/T = c/\lambda$.
c: the speed of light in a vacuum ($c = 3 \times 10^8$ m/s)

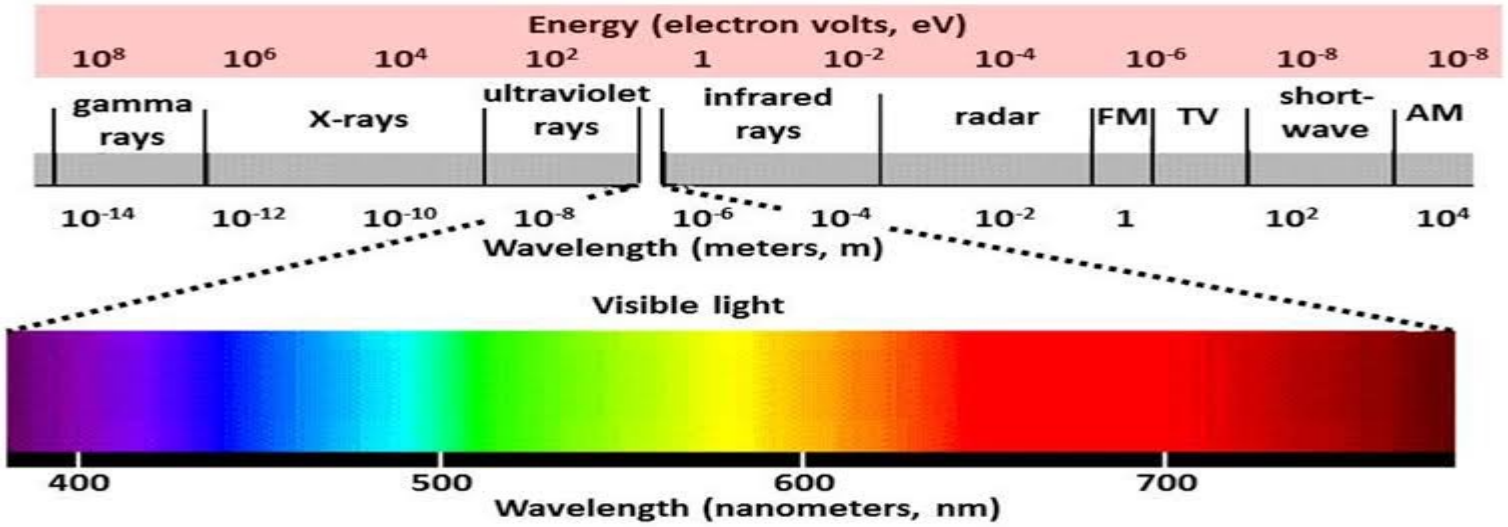
The wave number ν is the inverse of the wavelength: $\nu = 1/\lambda$ (Unit: nm^{-1}, cm^{-1}, m^{-1})

II.2.2. The Electromagnetic Spectrum

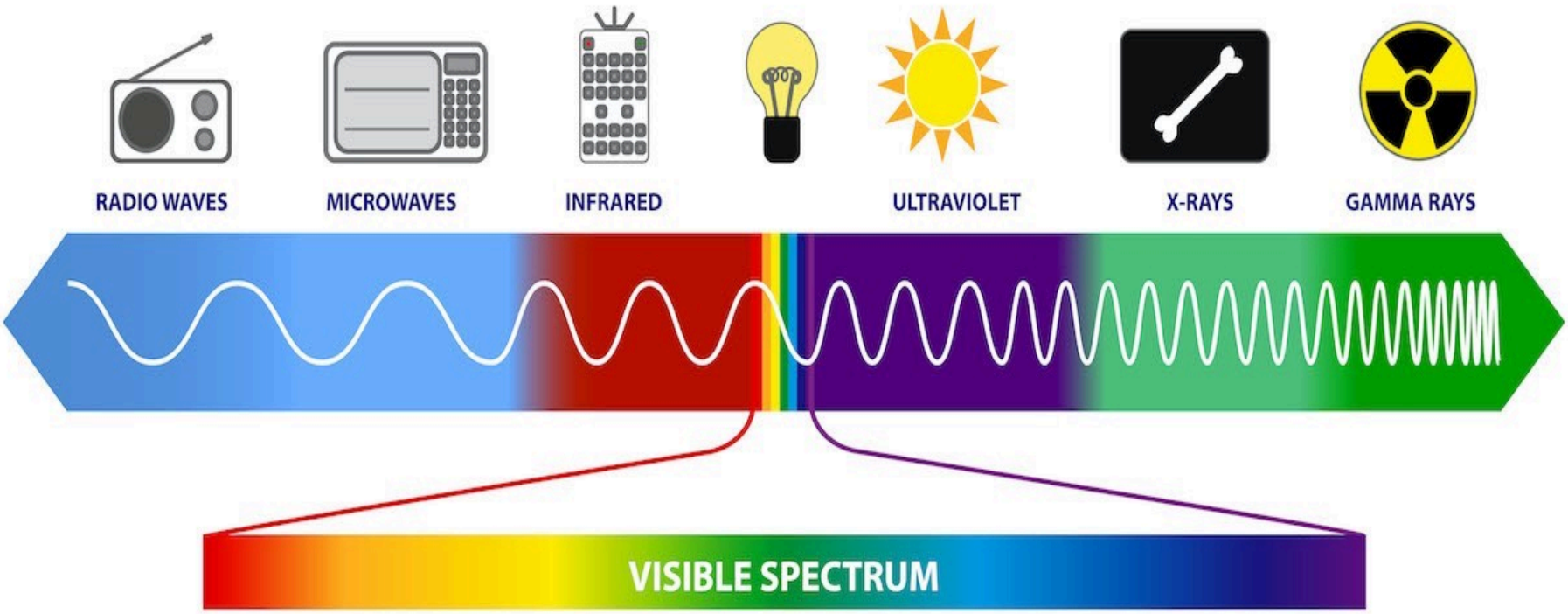
There are various types of electromagnetic radiation, ranging from gamma rays (γ)—which have very short wavelengths and high frequencies and result from the decay of radioactive substances—to waves with very long wavelengths and very low frequencies emitted by power transmission lines. This wide range of wavelengths and frequencies is called the electromagnetic spectrum .

The electromagnetic spectrum is largely invisible; to the naked eye, only a narrow portion located roughly in the middle of the spectrum is visible.

The Visible Spectrum. covers wavelengths ranging from approximately 390 nm—where violet appears just after ultraviolet—to approximately 760 nm, where red merges into infrared.



ELECTROMAGNETIC SPECTRUM



II.3. The Particle Nature of Light

Each component of white light is a form of light radiation characterized by a specific color (composed of an infinite number of colors: red, blue, violet, etc.), and each color corresponds to a specific energy, frequency, and wavelength. The particle nature of light radiation can be viewed as a collection of particles, each carrying a quantum of energy called a photon, whose value is proportional to its associated frequency.

$$E = h\nu = hc\lambda$$

Where h is Planck's constant: $h = 6.62 \times 10^{-34}$ J·s,
E is the light energy carried by the radiation in joules
, ν is the frequency of the radiation in (s^{-1}) or (Hz)
λ is the wavelength of the radiation in meters.

Wave-particle duality: Wave associated with the electron:
a. De Broglie's relation

Planck and Einstein had already demonstrated the dual nature—both wave-like and particle-like—
of light (wave-particle duality). In 1925, Louis de Broglie showed that to any moving material particle (corpuscle) of mass m, traveling at velocity
v, a guiding wave of wavelength λ can be associated such that: $\lambda = h/mv$. Between the
frequency ν of the associated wave and the energy E of the particle, the following relationship holds: $E = h\nu$
In the De Broglie model, the electron is described by a standing wave such that

$$2\pi r = n\lambda = \frac{nh}{mv}$$

In other words: the total length of the orbit is equal to an integer multiple of the wavelength of the De Broglie wave

II.4. Emission Spectrum of the Hydrogen Atom

There is a simple method for analyzing a radiation source: it is passed through a transparent prism, which separates the radiation into its characteristic wavelengths, forming a spectrum.

In the visible region, this spectrum can be seen directly on a graduated scale; in the ultraviolet or infrared regions, the wavelengths are detected electronically.

A spectroscope is used to perform this analysis. When the radiation source is white light, such as sunlight or light from incandescent solids (tungsten bulbs), the dispersion by the prism forms a continuous spectrum .

When the radiation results from electrical discharges in a tube containing a low-pressure gas (hydrogen, fluorescent tube, sodium or mercury lamp), the dispersion produces a discrete spectrum or line spectrum .

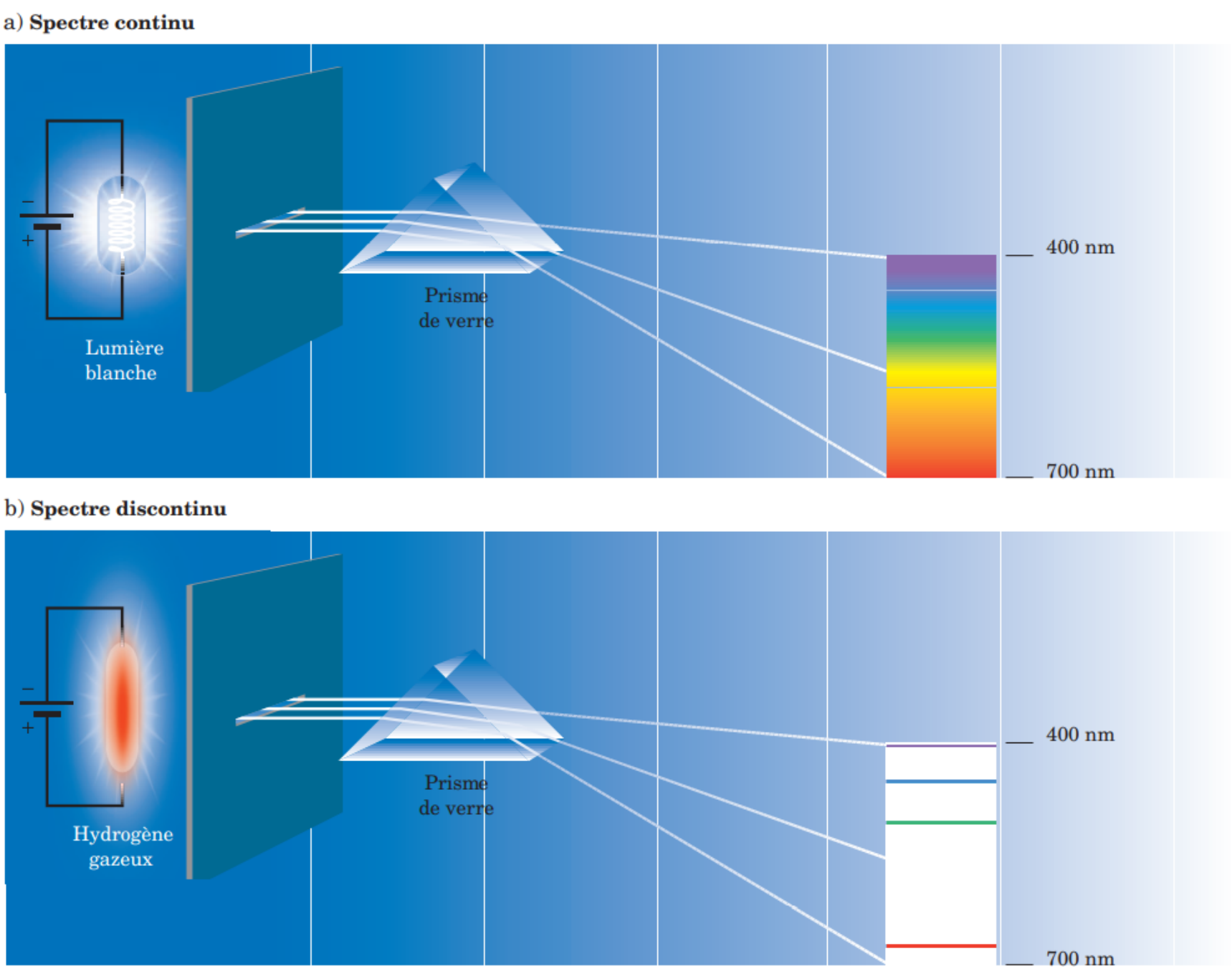


Figure : Continuous spectrum of white light and line spectrum of the hydrogen atom

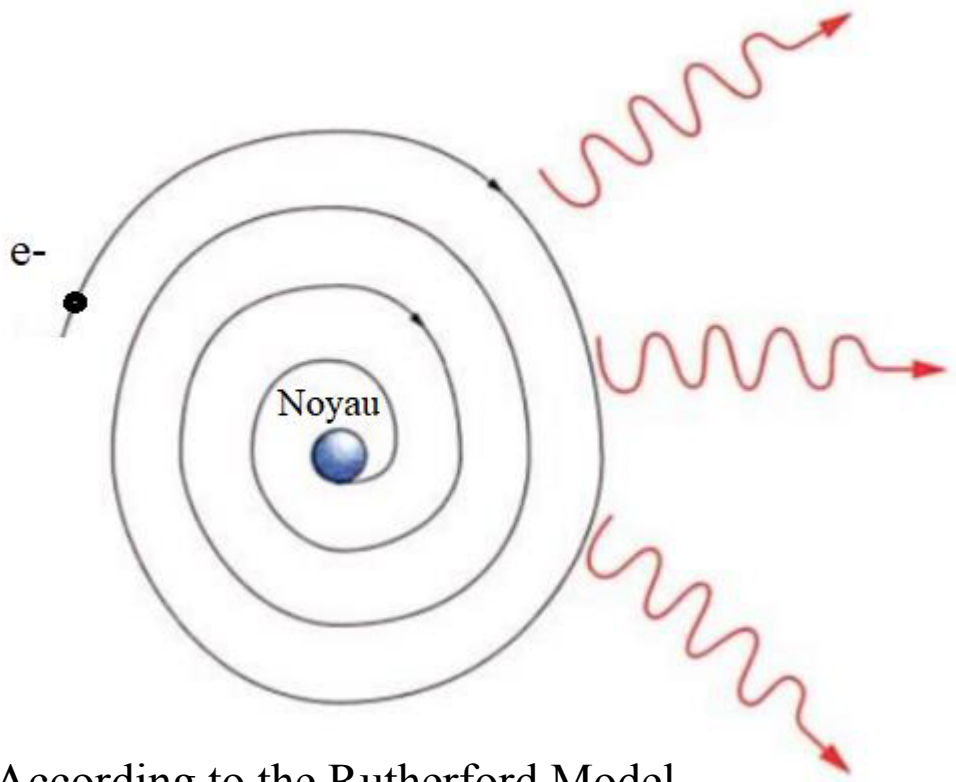
II.5. Bohr's Atomic Model – The Hydrogen Atom

In 1911, Rutherford described the atom as consisting of a dense nucleus around which electrons orbit, just as planets orbit the sun.

According to the laws of electrodynamics, the electron should emit electromagnetic radiation as it orbits the nucleus .

This radiation carries energy, which is lost during its orbit; the electron therefore moves closer and closer to the nucleus until it falls into it.

This presents a contradiction, since the electron does not fall—after all, the atom actually e



Electron Trajectory According to the Rutherford Model

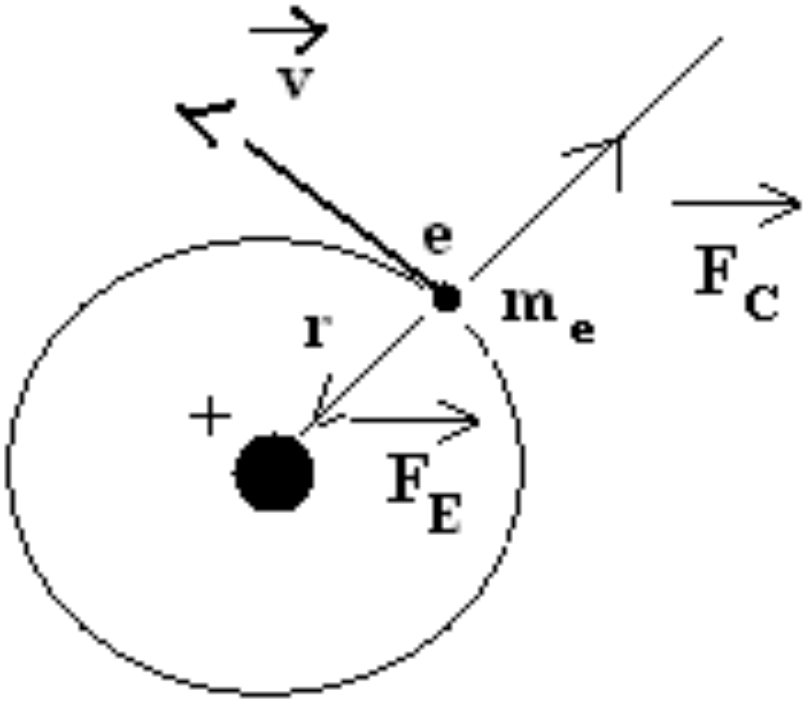
To resolve this contradiction, Bohr proposed a new atomic model in 1913 based on the quantization of energy. This model accounted for the structure of the hydrogen atom and its emission spectrum,. The characteristics of this model are summarized in three postulates:

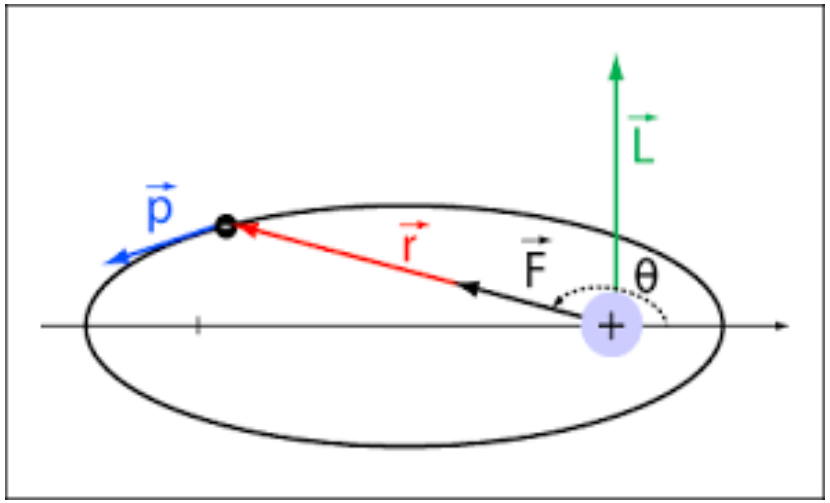
Bohr's First Postulate
The electron in a hydrogen atom moves around the nucleus in stationary circular orbits (of constant energy).

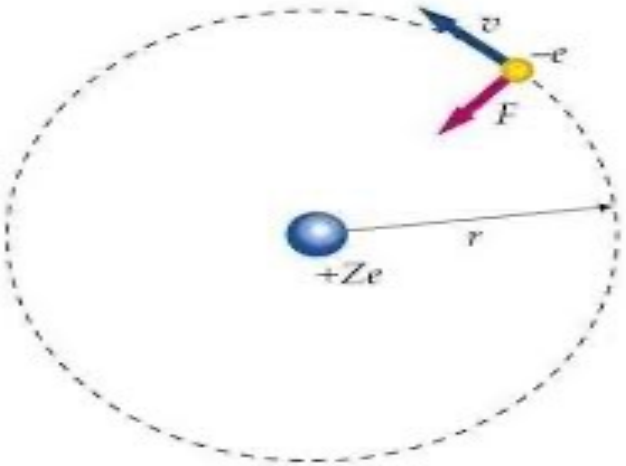
Bohr's Second Postulate The possible trajectories of electrons in the hydrogen atom are those for which the angular momentum **(mvr)** is an integer multiple of the quantity $h2\pi$, also called the quantum unit. Consequently, angular momentum is quantized as $mvr = nh2\pi.$

Where h is Planck's constant; m is the mass of the electron; and n is the principal quantum number.

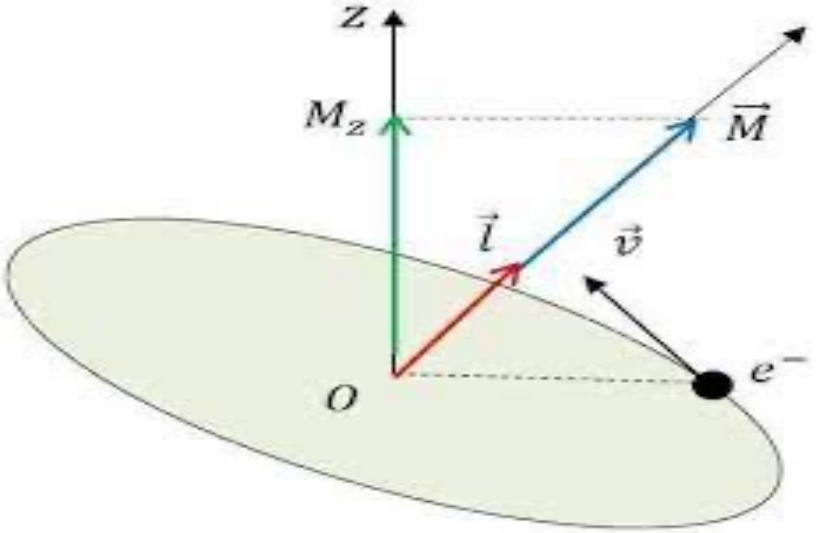
Bohr's Third Postulate
When the electron in a hydrogen atom transitions from one stationary orbit to another, it absorbs or emits electromagnetic radiation hν such that: **$\Delta E = |En1 - En2| = h\nu$**







$$F = \frac{kZe^2}{r^2} = \frac{mv^2}{r}$$



II.5.1. Radius of the Stationary Orbit

For the orbit to be stationary and stable, the centrifugal force acting on the electron must exactly balance the attractive force between the electron and the nucleus.

Centrifugal force: $Fc = mv^2/r$

Attractive force: $Fa = k(-e)(+e)/r^2 = -ke^2/r^2$ m: mass of the electron

v: velocity of the electron

r: distance between the electron and the nucleus (orbit radius)

e: charge of the electron

$k = 9 \times 10^9$ in the MKSA system

$||\vec{Fc}|| = ||\vec{Fa}|| \Rightarrow m\,v^2/r = ke^2/r^2 \Rightarrow mv^2 = ke^2/r$ (1)

According to Bohr's second postulate describing the quantization of angular momentum, we have:

$mvr = nh/2\pi \Rightarrow m^2v^2r^2 = (nh/2\pi)^2 \Rightarrow mv^2 = n^2h^2/(4\pi^2r^2m)$ (2)

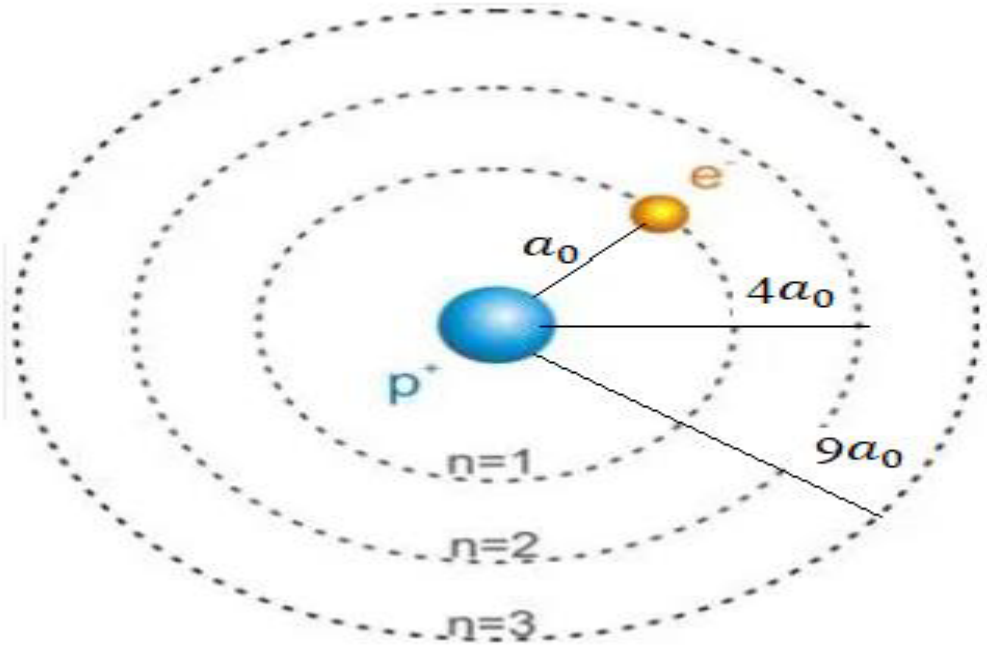
Combining equation (1) with equation (2) yields the expression for the orbital radius:

$ke^2r = n^2h^2/4\pi^2r2m \Rightarrow r = h^2/4k\pi^2me^2\ n^2$

We can deduce that the radius of the orbit is also quantized:

$$\mathbf{r_n = a_0n^2}$$

With $a_0 = h/24k\pi^2me^2 = 0.53$ Å (Å: Angstrom; 1 Å = 10^{-10} m). a_0 is called the radius of the first Bohr orbit.



Representation of the circular orbits of the hydrogen atom

The velocity of an electron in a stationary orbit is

$mvr = nh/2\pi \Longrightarrow v = nh/2\pi mr;$

substituting r with a0n2 $\longrightarrow$ $v = nh/2\pi m\ a0n2 \Longrightarrow v = h/2\pi m\ a0 \cdot 1n$

$\mathbf{vn} = \mathbf{v0n}$; $v_0 = h^2/\pi m\ a_0 = 2.18 \times 10^6$ m/s

II.5.2. Energy of an electron in a stationary orbit

$\overrightarrow{F_a} + \overrightarrow{F_c} = \vec{0}$ $\left\|\overrightarrow{F_a}\right\| = \left\|\overrightarrow{F_c}\right\|$ Avec $\left\|\overrightarrow{F_c}\right\| = \frac{m_e v^2}{r}$ et $\left\|\overrightarrow{F_a}\right\| = \frac{Ke^2}{r^2}$

Or $\left\|\overrightarrow{F_a}\right\| = \left\|\overrightarrow{F_c}\right\| \Longrightarrow \frac{m_e v^2}{r} = \frac{Ke^2}{r^2} \Longrightarrow m_e v^2 = \frac{Ke^2}{r}$ (1)

The total energy of the system is equal to the sum of its kinetic energy and potential energy

$$E_P = -\frac{K\ e^2}{r} \quad \text{et} \quad E_C = \frac{m_e v^2}{2}$$

$$E_P = \int_{\infty}^{r} F_a\, dr \Rightarrow E_P = -\frac{Ze^2}{4\pi\varepsilon_0 r} \text{ et } E_c = \frac{m_e v^2}{2} \text{ or } m_e v^2 = \frac{Ze^2}{4\pi\varepsilon_0 r} \Rightarrow E_c = -\frac{Ze^2}{8\pi\varepsilon_0 r}$$

$\boldsymbol{k}$ = 1 /4$\pi\varepsilon$0

ou ε0 = 8,85 × 10−12$C2N$−1m−2 (*permitivité du vide*)

$$m_e v^2 = \frac{Ke^2}{r} \Rightarrow \frac{m_e v^2}{2} = \frac{Ke^2}{2r}$$

$$E_T = -\frac{K\ e^2}{r} + \frac{Ke^2}{2r} = -\frac{Ke^2}{2r} \Rightarrow \boxed{E_T = -\frac{Ke^2}{2r}}$$

According to Bohr's second postulate, which describes the quantization of orbital angular momentum, we have:

$$m_e vr = n\frac{h}{2\pi} \Rightarrow m_e{}^2 v^2 r^2 = n^2 \frac{h^2}{4\pi^2} \Rightarrow m_e v^2 = n^2 \frac{h^2}{4m_e \pi^2 r^2}$$

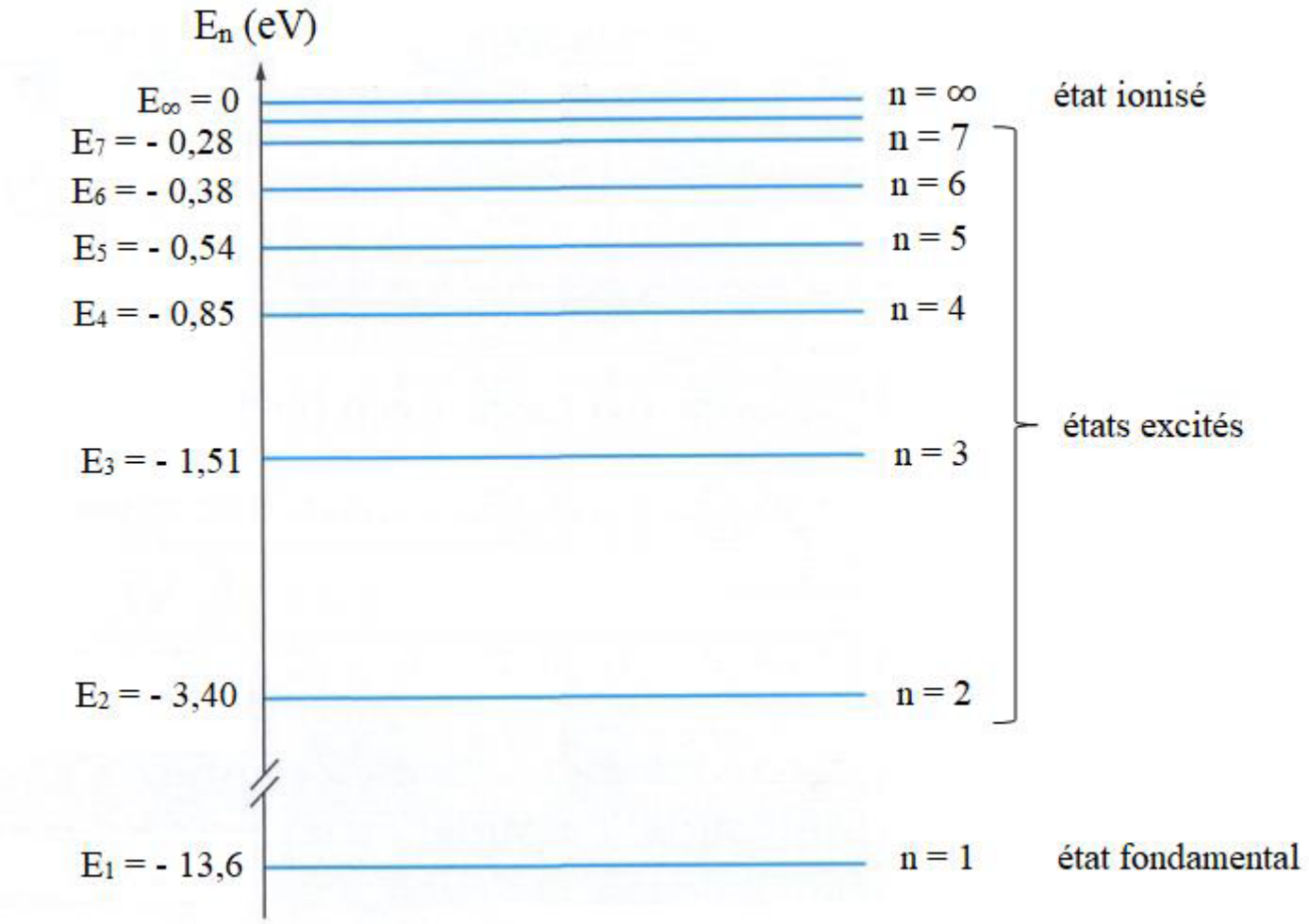
$$r = n^2 \frac{h^2}{4Km_e \pi^2 e^2}$$

Pour n=1 , r=0,529 Å, Le premier rayon de l'atome de Bohr

$$E_T = -\frac{Ke^2 4\pi^2 K m_e e^2}{2h^2}\frac{1}{n^2} = -\frac{2K^2\pi^2 m_e e^4}{h^2}\frac{1}{n^2}$$

$$E_T = -\frac{2K^2\pi^2 m_e e^4}{h^2}\frac{1}{n^2}$$

$$E_n = E_1 = -\frac{2K^2\pi^2 m_e e^4}{h^2}$$ = - 21.76 × 10^{-19} J = - 13.6 eV (this is the energyof the ground state of the hydrogen atom).



Energy Diagram of the Hydrogen Atom